

Assignment : 2

E.	Sepal Length	Sepal Width	distance	Rank
5.3	3.7	$\sqrt{(5.3-5.2)^2 + (3.7-3.1)^2} = 0.608$	9	
5.1	3.8	$\sqrt{(5.1-5.2)^2 + (3.8-3.1)^2} = 0.707$	6	
7.2	3.0	$\sqrt{(7.2-5.2)^2 + (3.0-3.1)^2} = 2.002$	13	
5.4	3.4	$\sqrt{(5.4-5.2)^2 + (3.4-3.1)^2} = 0.360$	2	
5.1	3.3	$\sqrt{(5.1-5.2)^2 + (3.3-3.1)^2} = 0.224$	1	
5.4	3.9	$\sqrt{(5.4-5.2)^2 + (3.9-3.1)^2} = 0.825$	8	
7.4	2.8	$\sqrt{(7.4-5.2)^2 + (2.8-3.1)^2} = 2.220$	15	
6.1	2.8	$\sqrt{(6.1-5.2)^2 + (2.8-3.1)^2} = 0.948$	10	
2.3	2.9	$\sqrt{(7.3-5.2)^2 + (2.9-3.1)^2} = 2.109$	14	
6.0	2.7	$\sqrt{(6.0-5.2)^2 + (2.7-3.1)^2} = 0.894$	9	
5.8	2.8	$\sqrt{(5.8-5.2)^2 + (2.8-3.1)^2} = 0.671$	5	
6.3	2.83	$\sqrt{(6.3-5.2)^2 + (2.3-3.1)^2} = 1.360$	12	
5.1	2.5	$\sqrt{(5.1-5.2)^2 + (2.5-3.1)^2} = 0.608$	3	
6.3	2.5	$\sqrt{(6.3-5.2)^2 + (2.5-3.1)^2} = 1.753$	11	
5.5	2.4	$\sqrt{(5.5-5.2)^2 + (2.4-3.1)^2} = 0.761$	7	

Considering $k = 3$

∴ Species : *Setosa*

2. $d_{11} = \sqrt{(0.1-0.2)^2 + (0.3-0.4)^2} = 0.141$ (closest)
 $d_{12} = \sqrt{(0.5-0.2)^2 + (0.8-0.4)^2} = 0.5$
 $d_{13} = \sqrt{(0.3-0.2)^2 + (0.7-0.4)^2} = 0.22$
 $d_{22} = \sqrt{(0.4-0.2)^2 + (0.7-0.4)^2} = 0.36$

∴ Winning Node : $W_{11} = [0.1, 0.3]$

3.

$$d_{11} = \sqrt{(0.2-0.1)^2 + (0.4-0.3)^2 + (0.7-0.6)^2} = 0.173$$

$$d_{12} = \sqrt{(0.2-0.5)^2 + (0.4-0.8)^2 + (0.7-0.2)^2} = 0.707$$

$$d_{13} = \sqrt{(0.2-0.7)^2 + (0.4-0.5)^2 + (0.7-0.9)^2} = 0.724$$

$$d_{21} = \sqrt{(0.2-0.3)^2 + (0.4-0.2)^2 + (0.7-0.8)^2} = 0.245$$

$$d_{22} = \sqrt{(0.2-0.6)^2 + (0.4-0.1)^2 + (0.7-0.4)^2} = 0.583$$

$$d_{23} = \sqrt{(0.2-0.7)^2 + (0.4-0.3)^2 + (0.7-0.3)^2} = 0.648$$

$$d_{31} = \sqrt{(0.2-0.4)^2 + (0.4-0.7)^2 + (0.7-0.5)^2} = 0.412$$

$$d_{32} = \sqrt{(0.2-0.3)^2 + (0.4-0.3)^2 + (0.7-0.2)^2} = 0.173$$

$$d_{33} = \sqrt{(0.2-0.1)^2 + (0.4-0.4)^2 + (0.7-0.7)^2} = 0.509$$

$\therefore$ Winning Node : $W_{32} [0.3, 0.3, 0.6]$

4.

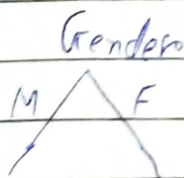
[Bus, Train, car]

[4, 3, 3]

$$E = -\frac{4}{10} \log_2 \left(\frac{4}{10} \right) - \frac{3}{10} \log_2 \left(\frac{3}{10} \right) - \frac{3}{10} \log_2 \left(\frac{3}{10} \right)$$

$$= 0.528 + 0.521 + 0.521$$

$$= 1.57$$



[3, 1, 1] [1, 2, 2]

$$E_H = -\frac{3}{5} \log_2 \left(\frac{3}{5} \right) - \frac{1}{5} \log_2 \left(\frac{1}{5} \right) - \frac{1}{5} \log_2 \left(\frac{1}{5} \right)$$

$$= 0.492 + 0.469 + 0.469$$

$$= 1.37$$

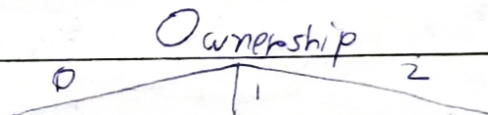
$$E_F = -\frac{1}{5} \log_2 \left(\frac{1}{5}\right) - \frac{2}{5} \log_2 \left(\frac{2}{5}\right) - \frac{2}{5} \log_2 \left(\frac{2}{5}\right)$$

$$= 0.464 + 0.528 + 0.528$$

$$= 1.52$$

$$\text{Gain} = 1.57 - \frac{5}{10} (1.37) - \frac{5}{10} (1.52)$$

$$= 0.125$$



$$[2, 1, 0] \quad [2, 2, 1] \quad [0, 0, 2]$$

$$E_0 = -\frac{2}{3} \log_2 \left(\frac{2}{3}\right) - \frac{1}{3} \log_2 \left(\frac{1}{3}\right)$$

$$= 0.389 + 0.528$$

$$= 0.917$$

$$E_1 = -\frac{2}{5} \log_2 \left(\frac{2}{5}\right) - \frac{2}{5} \log_2 \left(\frac{2}{5}\right) - \frac{1}{5} \log_2 \left(\frac{1}{5}\right)$$

$$= 0.528 + 0.528 + 0.464$$

$$= 1.52$$

$$E_2 = 0$$

$$\text{Gain} = 1.57 - \frac{3}{10} (0.917) - \frac{5}{10} (1.52)$$

$$= 0.535$$



$$[4, 1, 0] \quad [0, 2, 0] \quad [0, 0, 3]$$

$$E_c = \frac{-4}{5} \log_2 \left(\frac{4}{5} \right) - \frac{1}{5} \log_2 \left(\frac{1}{5} \right)$$

$$= 0.757 + 0.469$$

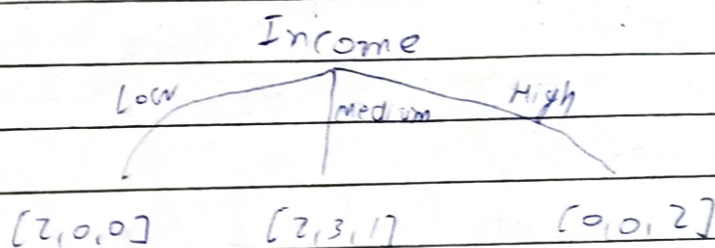
$$= 0.721$$

$$E_s = 0$$

$$E_e = 0$$

$$\text{Gain} = 1.57 - \frac{5}{10} (0.721)$$

$$= 1.209$$



$$E_L = 0$$

$$E_M = \frac{-2}{6} \log_2 \left(\frac{2}{6} \right) - \frac{3}{6} \log_2 \left(\frac{3}{6} \right) - \frac{1}{6} \log_2 \left(\frac{1}{6} \right)$$

$$= 0.528 + 0.5 + 0.431$$

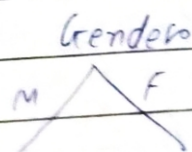
$$= 1.459$$

$$E_H = 0$$

$$\text{Gain} = 1.57 - \frac{6}{10} (1.459)$$

$$= 0.6946$$

∴ Travel Cost is root node.



$$[3, 0, 0] \quad [1, 1, 0]$$

$$E_M = 0$$

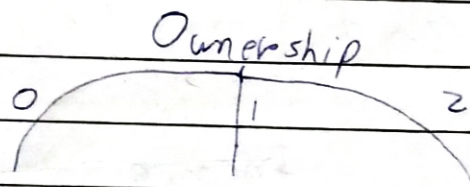
$$E_F = -\frac{1}{2} \log_2 (1/2) - \frac{1}{2} \log_2 (1/2)$$

$$= 0.5 + 0.5$$

$$= 1$$

$$Gain = \frac{0.721}{5} - \frac{2}{5} (1)$$

$$= \cancel{1.57} 0.321$$



$$[2, 0, 0] \quad [2, 1, 0] \quad [0, 0, 0]$$

$$E_0 = 0$$

$$E_1 = -\frac{2}{3} \log_2 (2/3) - \frac{1}{3} \log_2 (1/3)$$

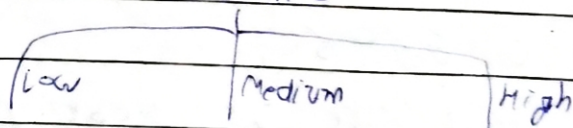
$$= 0.389 + 0.528$$

$$= 0.917$$

$$Gain = \frac{0.721}{5} - \frac{3}{5} (0.917)$$

$$= \cancel{1.018} 0.171$$

Income



$$[2, 0, 0] \quad [2, 1, 0] \quad [0, 0, 0]$$

$$E_L = 0$$

$$E_M = -\frac{2}{3} \log_2 (2/3) - \frac{1}{3} \log_2 (1/3)$$

$$= 0.389 + 0.528$$

$$= 0.917$$

$$E_H = 0$$

$$\text{Gain} = 0.721 - \frac{3}{5} (0.917)$$

$$= 0.171$$

∴ Next best for cheap cost is Gender

